Julio de Mello Bezerra

Mathematics PhD | Cryptography & Cryptanalysis

Regensburg, Germany • mellojulio96@gmail.com • +49 176 48727782

Profile

PhD Mathematician transitioning from pure mathematics into applied cryptography, bringing a strong theoretical foundation in classical and post-quantum encryption. Proven ability to translate advanced mathematical literature into functional code, with practical experience implementing attacks and algorithms in Python and SageMath. Track record of analytical problem-solving, evidenced by academic history, cryptographic research projects and CTF challenges.

Language Skills

Languages: English (Fluent), Portuguese (Native), German (Working Proficiency), Spanish (Basic Proficiency)

Programming: Python, SageMath, LaTeX

Cryptographic Research & Projects

Yukon CurveSmash Project (yukon.org/curvesmash) **2026 – Present**
Judge & Cryptanalysis Researcher *Remote*

- Evaluated and advanced state-of-the-art elliptic curve cryptanalysis in a project combining modern AI capabilities with human mathematical expertise, resulting in a co-authored article (*Beyond the Cheon-Kozaki-Kutsuma-Matsuo attack*) analyzing improved attack parameters.
- Investigating pairing inversion security on modern pairing-friendly curves, with an upcoming technical article in preparation, by applying the Special Tower Number Field Sieve (STNFS) attack to BLS12-381.

Applied & Theoretical Cryptography **2026**
Projects & Study *Independent*

- **Algorithm Implementation:** Authored a technical article and Python/SageMath implementation demonstrating how to construct strong pseudoprimes that bypass primality tests, by translating classical cryptography papers into optimized executable code.
- **CTF Challenges:** Solved a wide range of Capture-the-Flag challenges on cryptohack.org (reaching the top 1% of users) by implementing practical attacks across most standard symmetric and asymmetric schemes.
- **Post-Quantum:** Studied general post-quantum protocols and isogeny-based hardness assumptions as a participant in the QuSoft reading group on *Quantum Algorithms for Isogenies* (Spring 2021).

Professional Experience

University of Regensburg **Oct 2020 – Jul 2026**
PhD Researcher & Lecturer *Regensburg, Germany*

- **Research:** Conducted independent research in number theory and arithmetic geometry resulting in a *magna cum laude* doctoral thesis and a preprint, alongside collaborative research in advanced research seminars (*Oberseminare*).
- **Teaching:** Extensive teaching experience across 8 semesters in both English and German, in general undergraduate math courses as well as advanced seminars in cryptography-relevant topics such as Elliptic Curves and Lattices.
- **Conferences:** Co-organized a workshop on Eisenstein series and equivariant cohomology, delivered research talks in Germany, France, and the UK, and participated in multiple conferences across Europe.

Education

University of Regensburg **Apr 2026**
Doktor der Naturwissenschaften (PhD in Mathematics) – magna cum laude *Regensburg, Germany*

- **Thesis:** *Shintani cocycle decomposition of topological polylogarithms*

University of Regensburg & Leiden University **Aug 2020**
ALGANT Master in Mathematics *Germany & Netherlands*

- Scholarship recipient in joint international master's programme focused on algebra, geometry, and number theory.

Universidade Federal do Rio de Janeiro (UFRJ) **Jan 2018**
Bachelor in Pure Mathematics *Rio de Janeiro, Brazil*

Selected Publications & Awards

- **Mathematics Preprint:** J. de Mello Bezerra, "Shintani cocycle decomposition of topological polylogarithms," arXiv:2610.32363 (2026).
- **Award:** Silver Medal (30th out of 9,531 participants), Brazilian Computer Science Olympics (2010).